

B.TECH. COMPUTER SCIENCE AND ENGINEERING (AI& ML)
Deenbandhu Chhotu Ram University of Science & Technology Murthal, Sonapat, Haryana
Semester – III (Effective from Session 2025-26)

Course Code: PCCSE203	Minor	TCA	TES	PCA	PES	Total
Title of the Course: OOP USING JAVA	15	10	50	15	10	100
L-T-P: 3-0-2						
Credits: 4	Duration of Exam: 3 Hours					
Brief Write-Up About the Course (100-150 words)						
This course introduces students to object-oriented programming using Java, a powerful, platform-independent language widely used in software development. Beginning with Java fundamentals such as data types, operators, control structures, and classes, students gain a solid grounding in writing and executing standalone Java programs. The course emphasizes core object-oriented principles, including inheritance, polymorphism, abstraction, and interfaces, equipping students with reusable and modular design techniques. Students also explore the Java Collection Framework, file handling using streams, and robust exception handling mechanisms. Advanced topics include GUI programming with Swing, event handling using the delegation model, and database connectivity through JDBC. By the end of the course, students will be able to develop interactive, data-driven Java applications and understand the architecture and best practices of modern Java programming.						
Course Objective (CO):						
1. To provide object-oriented concepts through which robust, secured and reusable software can be developed. 2. To understand the principles of inheritance and polymorphism and demonstrate how they relate to the design of abstract classes. 3. To understand the concepts of file handling and exception handling. 4. To understand the design of Graphical User Interface using applets and swing controls.						
UNIT - I						
Java Programming- History of Java, comments, Data types, Variables, Constants, Scope and Lifetime of variables, Operators, Type conversion and casting, Enumerated types, Control flow-block scope, conditional statements, loops, break and continue statements, arrays, simple java stand alone programs, class, object, and its methods constructors, methods, static fields and methods, access control, this reference, overloading constructors, recursion, exploring string class, garbage collection.						
UNIT - II						
Inheritance – Inheritance types, super keyword, preventing inheritance: final classes and methods. Polymorphism – method overloading and method overriding, abstract classes and methods.						
Interfaces- Interfaces Vs Abstract classes, defining an interface, implement interfaces, accessing implementations through interface references, extending interface, inner class. Packages- Defining, creating and accessing a package, importing packages.						
UNIT-III						
Collection Framework in Java – Introduction to java collections, Overview of java collection framework, Commonly used collection classes- Array List, Vector, Hash table, Stack, Lambda Expressions.						
Files- Streams- Byte streams, Character streams, Text input/output, Binary input/output, File management using File class.						
Exception handling -Benefits of exception handling, the classification of exceptions- exception hierarchy, checked exceptions and unchecked exceptions, usage of try, catch, throw, throws and finally, creating own exception subclasses.						

UNIT-IV

GUI Programming with Swing - The AWT class hierarchy, Introduction to Swing, Swing Vs AWT, Hierarchy for Swing components, Overview of some Swing components – JButton, JLabel, JTextField, JTextArea, simple Swing applications, Layout management – Layout manager types – border, grid and flow.

Event Handling- Events, Event sources, Event classes, Event Listeners, Delegation event model, Examples: Handling Mouse and Key events, Adapter classes.

Connecting to Database – JDBC Type 1 to 4 drivers, Connecting to a database, querying a database and processing the results, updating data with JDBC.

Text books:

1. Java Fundamentals—A Comprehensive Introduction, Herbert Schildt and Dale Skrien, TMH.
2. Core Java: An Integrated Approach – Dr R Nageswara Rao

Reference books:

1. Java for Programmers, P.J.Deitel and H.M.Deitel, PEA (or) Java: How to Program , P.J.Deitel and H.M.Deitel, PHI
2. Object Oriented Programming through Java, P.Radha Krishna, Universities Press.
3. Programming in Java, S. Malhotra and S. Choudhary, Oxford Universities Press.
4. Design Patterns Erich Gamma, Richard Helm, Ralph Johnson and John Vlissides.

Course Outcome(CO):

Upon successful completion of the course, students will demonstrate the ability to:

1. A competence to design, write, compile, test and execute straight forward programs using JAVA.
2. Be able to implement basic principles of object-oriented programming.
3. Be able to deal with file handling and exceptional handling.
4. Be able to create effective website using AWT and swings.

List of Programs:

1. Write a java program to find the Fibonacci series using recursive and non recursive functions.
2. Write a java program to multiply two given matrices.
3. Write a java program for Method overloading and Constructor overloading.
4. Write a java program to display the employee details using class.
5. Write a java program that checks whether a given string is palindrome or not
6. a. Write a java program to represent Abstract class with example.
b. Write a java program to implement Interface using extends keyword.
7. a. Write a java program to create inner classes.
b. Write a java program to create user defined package.
8. Write a java program for creating multiple catch blocks.
9. a. Write a java program to display File class properties.
b. Write a java program to represent ArrayList class.
c. Write a Java program loads phone no, name from a text file using hash table.
10. Write an applet program that displays a simple message.
11. Write a Java program to compute factorial value using Applet.
12. Write a program for passing parameters using Applet.
13. Write a simple program using Swing components – JButton, JLabel, JTextField, JTextArea.
14. Write a java program that works as a simple calculator. Use a Grid Layout to arrange Buttons for digits and for the + - * % operations. Add a text field to display the result
15. Write a java program for handling Mouse events.
16. Write a java program for handling Key events.
17. Write a java program to connect to a database using JDBC and insert values into it.
18. Write a java program to connect to a database using JDBC and delete values from it.